



SQL Assignment

You are provided with two tables:

1. **Employees :**
| EmployeeID | Name | Department | Salary | JoinDate |
2. **Projects :**
| ProjectID | ProjectName | Department | Budget |

DataSets:

<https://drive.google.com/drive/folders/16fkU2DQumhYik0Ft6FG07QT32E3DpI3K?usp=sharing>

Tasks

Task 1: Create the Tables and Insert Data

Write SQL scripts to create the `Employees` and `Projects` tables and insert the sample data.

Task 2: Retrieve All Employees in the IT Department

Write a query to list all employees who work in the IT department.

Solution:

```
SELECT *  
FROM Employees  
WHERE Department = 'IT';
```

Task 3: Find Employees Who Joined After 2020

Retrieve the names of employees who joined after January 1, 2020.

Solution:

```
SELECT Name  
FROM Employees  
WHERE JoinDate > '2020-01-01';
```

Task 4: Calculate Total Budget by Department

Write a query to calculate the total project budget for each department.

**Solution:**

```
SELECT Department, SUM(Budget) AS TotalBudget
FROM Projects
GROUP BY Department;
```

Task 5: List Employees with a Salary Above the Average

Retrieve the details of employees whose salaries are above the average salary of all employees.

Solution:

```
SELECT *
FROM Employees
WHERE Salary > (SELECT AVG(Salary) FROM Employees);
```

Task 6: Find Departments Without Projects

Write a query to find departments in the `Employees` table that have no associated projects in the `Projects` table.

Solution:

```
SELECT DISTINCT Department
FROM Employees
WHERE Department NOT IN (SELECT DISTINCT Department FROM Projects);
```

Task 7: Add a Column for Performance Rating

Modify the `Employees` table to add a column `PerformanceRating` (default value as `NULL`), and update it as follows:

- HR employees: Rating = 4
- IT employees: Rating = 5

Solution:

```
ALTER TABLE Employees ADD PerformanceRating INT;

UPDATE Employees
SET PerformanceRating = 4
WHERE Department = 'HR';

UPDATE Employees
SET PerformanceRating = 5
WHERE Department = 'IT';
```



Task 8: Display Employees and Their Project Details

Join the `Employees` and `Projects` tables to display each employee's name, department, and the project(s) associated with their department.

Solution:

```
SELECT e.Name, e.Department, p.ProjectName, p.Budget
FROM Employees e
LEFT JOIN Projects p ON e.Department = p.Department;
```

Additional Challenge

Task 9: Create a View for High Earners Create a view `HighEarners` that contains the details of employees earning more than 60,000.

Solution:

```
CREATE VIEW HighEarners AS
SELECT *
FROM Employees
WHERE Salary > 60000;
```